An Application of Newsvendor Model on Pipe Stringing Operations of a Gas Pipeline Construction Project

Yuqiang Wei, Chenggong Feng
China Petroleum Pipeline Engineering Co., Ltd., Langfang, China

Shunichi Ohmori
Graduate School of Creative Science and Engineering, Waseda University, Tokyo 169-8050, Japan.

Arnat Watanasungsuit
Hydrocarbon Solution (Thailand) Company Limited, Thawi Wattana District, Bangkok 10170, Thailand.

Choosak Pornsing, Kawinthorn Saichareon*
Department of Industrial Engineering and Management, Faculty of Engineering and Industrial Technology, Silpakorn University, Nakhon Pathom, 73000 Thailand.

ABSTRACT

With the increasingly fierce market competition, enterprises must maintain their invincible position through continuous improvement, innovation, and optimization of their operations to save cost and maximize profit. In this article, the authors introduce a resource allocation technique for a long-distance gas pipeline construction project. Through the introduction of the newsvendor model, the optimal resource allocation, workforce, and equipment, on the pipeline stringing operations, which makes the welding crew perform its functions are the crucial tactic of the operation manager. The idea of this study is to adopt the concept of the newsvendor model for trading between overstock cost and understock cost of gas pipe transportation management. The result of applying has shown that the model provided the optimal control of the pipeline stringing operation in which concern to the uncertainty of relevant factors. In this case study, the total cost is $760.13 per day compared to the total cost of the traditional method, delivering the pipes at an average quantity, which is $1,587.22 per day. As a result, the newsvendor model could save the company money by 52.11% on supply chain and logistics operations. Furthermore, the newsvendor model does not only provide a simple optimization technique but also shows a guide to businesses to think about applying other optimization tools to their business processes and manufacturing processes.

CCS CONCEPTS

• **Mathematics of computing** → Hypothesis testing and confidence interval computation.

KEYWORDS

newsvendor model, base-stock policy, supply chain and logistics operations, gas pipeline, petroleum industry

*Corresponding author: kawintho@gmail.com

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1 INTRODUCTION

In a gas pipeline construction project, the schedule management of pipe fabrication is regarded as a key performance indicator of the construction project. It relates to all activities in the project; material acquisition, welding and fabricating, stocking, transportation, pipe stringing, and so on. Therefore, all activities must be synchronized as supply chain management to make the whole chain works efficiently.

The pipe stringing activity, as the predecessor of pipe welding activity, is transporting the pipe from the stockyard to the site where the right of way has been cleaned and graded for the supply of pipeline prefabrication. Each time when the welding team mobilized to that site, pipes have to be ready for welding. However, how many resources should be invested in pipe stringing activity to satisfy the demand for pipe welding shall be considered.

Traditionally, the demand of pipe supply is calculated as per the required rate of pipe welding that is calculated with average values on a daily basis and the according resources will be invested. Practically, the number of welding joints completed by welding teams was not constant. Sometimes, the rate of welding would be higher on a good weather day or lower on a bad weather day. Consequently, in case that pipes were not delivered on-site, there would be no work to do for the welding team and the team will standby on-site without benefit, but the maintenance fee has to be paid before welding mobilization like land acquisition if pipes were delivered on-site but welding couldn't be completed timely. Hence, this method may have the backward that the actual demand couldn't be matched to maximize the benefit.

It has been observed that there was a relationship of supply chain between pipe stringing and welding, thus, the theory of inventory, named newsvendors model, would be considered to apply for calculating the demand of pipe supply on-site to minimize the cost and maximize the benefits so that the resource investment for pipe string activity could be considered accordingly.

As a result, the newsvendor model will be applied to calculate the demand of pipe supply for the pipe string activity in the following parts to find out the advantages of the method compared with the traditional method.

2 LITERATURE REVIEW

The mathematical analysis of the newsvendor problem was originated by Arrow et al. in 1951 [1], though some of the ideas were much older: Edgeworth (1888) used newsvendor type logic to determine the amount of cash to keep on hand at a bank to satisfy random withdrawals by patrons [2]. Morse and Kimball (1951) introduced the name "newsboy problem" [3], and Goker & Porteus (2008) cited Matt Sobel as proposing the gender-neutral term "newsvendor problem" [4].

Normally, h is used to represent the holding cost: the cost per unit of having too much inventory on hand. In the newsvendor problem, this typically consists of the purchase cost of the unit, minus any salvage value, but may include other costs, such as processing costs. (Since inventory cannot be carried to the next period, this cost is not technically a holding cost, though we will refer to it that way anyway.) Similarly, p represents the stockout cost: the cost per unit of having too little inventory, consisting of lost profit and loss of goodwill costs. The holding cost is the cost per unit of positive ending inventory, while the stockout cost is the cost per unit of negative ending inventory. The costs h and p are sometimes referred to as overage and underage costs, respectively (and some authors denote them C_o and C_u). We can assume that the purchase cost is included in h and that its negative is included in p, and therefore, we assume that the ordering cost is zero. D is a random variable that represents the demand per period, and the goal is to determine the base-stock level S to minimize the expected cost in the single period. The strategy for solving this problem is first to develop an expression for the cost as a function of d (the observed demand) and S (called $g(S, d)$); then to determine the expected cost $ED[g(S, d)]$ (called $g(S)$); and then to determine S to minimize $g(S)$.

Let $I(S; d) = (S - d)^+$ and $B(S; d) = (d - S)^+$ be the on-hand inventory and backorders respectively, at the end of the period if the firm orders up to S and sees a demand of d units. The cost for an observed demand of d is,

$$g(S,\ d) = h * I(S,\ d) + p * B(S,\ d) = h * (S - d)^+ + p * (d - S)^+ \tag{1}$$

The optimal base-stock level for a single-period model with no fixed costs (the newsvendor model) is given by and the expected cost $g(S^*)$ would be,

$$g\left(S^*\right) = (h + p)\,\phi\left(Z_\alpha\right)\sigma \tag{2}$$

where $Z_\alpha = \Phi - 1[p/(p + h)]$, Φ is the cumulative density function (CDF) of standardized normal distribution function, and ϕ is the probability density function (PDF) of standardized normal distribution function [5].

Olivares et al. (2008) [6] extended the newsvendor model to cope with unobservable cost parameters using an estimation framework. The framework is based on a statistical method, the structural model, that gives consistent estimations. The proposed approach was applied to manage the capacity of the operating rooms in a hospital. The model suggested that the hospital over accented on the cost of idle capacity than on the costs of schedule overrun. Herbon and Hadas (2015) [7] adopt the newsvendor model to solve the required capacity and the optimal frequency of bus routes. The overcrowding of passengers is the understock cost and the empty seats of a bus are the overstock cost. The interesting thing about this study is the duration of the period time is also a decision variable. The authors proposed an algorithm to solve the problem efficiently. It also solves a non-defined distribution stochastic demand. The results of the numerical computation showed that the model and the approach yielded optimal operator costs and could decrease the waiting time of the passengers simultaneously. Meza (2016) [8] systematically reviewed and find out the opportunities to apply the tools in supply chain management to the construction sites. The information was extracted from firms in Europe and United States. The study claimed that the newsvendor model could help a construction firm to control its project cost. The model could reveal an optimal point of cost and availability of supply. This model is also deployed in deciding on bidding and budgeting of the construction projects. Up to this point, our study is not a sophisticated study project. Nevertheless, it shows the managers in construction projects to apply a simple but powerful optimization tool to shape their business in an efficient way. Alwan et al. (2016) [9] developed the classic newsvendor model to cope with multi-period demand. The obstacle was the assumption that demand in each period were independence. The authors made use of correlated demands in the model. It showed that the modified newsvendor model with correlated demands under the period-to-period planning circumstance worked well with small mean square error.

3 RESEARCH METHODOLOGY

Normally, a certain quantity of pipes was welded by one team per day perfectly matched; the project will get the benefit from the welded pipes. However, in the case of pipes were not delivered on-site, there would be no work to do for the welding team; accordingly, all costs such as the rent fee, salaries, and other allowances of the team have to be paid as standby cost. On the other hand, pipes could be delivered to the site as much as possible but the maintenance fee has to be paid before welding team mobilization like safety guards and land acquisition to stock the pipe on site.

As per the characteristics of the newsvendor model, it follows the assumptions of the Base-stock model as below:

- Pipe supply can be analyzed individually;
- Demands occur once at a time (no batch orders) on daily basis;
- No stockout situation;
- Replenishment lead times are fixed and known;
- Replenishments are ordered once a time;
- Demand is modeled by a continuous probability distribution;

According to the characteristics of pipe supply during pipe string activity, the demand of pipe supply could be analyzed individually and on daily basis. Stockout situation will not happen for continuous construction and the order and supply will be on daily basis. The distribution of the demand will be verified in the following part as well as the penalty cost and holding cost. If the demand can be modeled by a continuous probability distribution and the penalty

Table 1: The observed data of pipe welding rate

Day	Quantity	Day	Quantity	Day	Quantity	Day	Quantity
1	16	32	23	63	12	94	17
2	16	33	19	64	13	95	18
3	14	34	24	65	14	96	20
4	6	35	28	66	7	97	21
5	16	36	22	67	7	98	21
6	18	37	23	68	6	99	14
7	18	38	18	69	11	100	20
8	18	39	21	70	11	101	21
9	24	40	13	71	15	102	23
10	23	41	18	72	14	103	18
11	26	42	13	73	17	104	24
12	24	43	20	74	15	105	24
13	24	44	19	75	18	106	26
14	21	45	18	76	19	107	21
15	25	46	18	77	15	108	23
16	18	47	18	78	17	109	14
17	25	48	19	79	20	110	24
18	21	49	22	80	13	111	24
19	22	50	17	81	22	112	29
20	22	51	18	82	17	113	26
21	22	52	21	83	20	114	29
22	24	53	18	84	9	115	23
23	22	54	14	85	9	116	31
24	20	55	19	86	16	117	30
25	20	56	25	87	15	118	25
26	18	57	21	88	18	119	22
27	20	58	22	89	16	120	28
28	18	59	24	90	11	121	29
29	22	60	18	91	25	122	30
30	17	61	17	92	19	123	26
31	25	62	12	93	13	124	23

cost and holding cost are constant, then, the newsvendor model could be applied for analysis.

4 RESULTS AND ANALYSIS

To find out the parameters from pipe string activity that may match with the model, a pilot survey was made to collect the data per each parameter for calculation. the welding rate that proceeded by the welding team had been collected in the duration of continuous 124 days. the details are shown in Table 1

From Table 1, the quantity of welded pipes each working day was collected. It was 124 consecutive days of operation. We needed to verify the data with the boxplot technique to identify unusual data. The result of the boxplot technique is shown in Fig. 1. It was found that 2 data were unusual, day 4 and day 68. This might deteriorate the analysis. Thus, we would like to remove these numbers from the data.

Then, we made use of the Anderson-Darling test for normality testing. In this case, we had no information on population mean and population variance. By setting $\alpha = 0.05$, we concluded that there was not sufficient evidence to reject the null hypothesis. As a result, the data were distributed as a normal distribution. Furthermore, the mean was 19.61 and its standard deviation was 4.957.

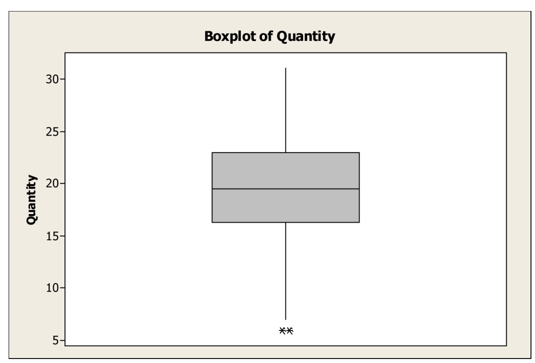

Figure 1: Boxplot of the observed data

Next, the holding cost and penalty cost for this study were described as below:

Holding cost = Maintenance cost + Risk cost

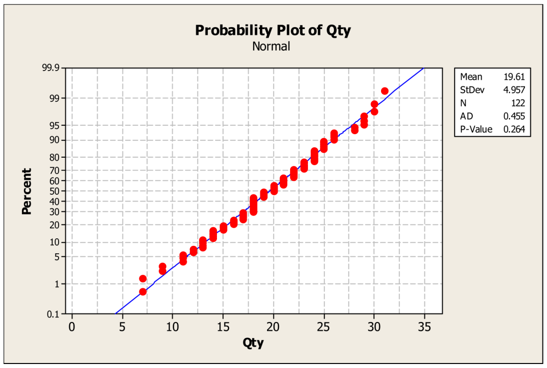

Figure 2: Normal probability plot

Where the maintenance cost includes safety barriers cost, warning signs cost, security cost, grounding cost, inspection cost, community relation cost, etc. The risk cost will emerge when pipes have been stolen, burned, or damaged, the strung pipes may also cause safety accidents, such as someone hitting the pipe and getting hurt.

Penalty cost = Profit loss + Standby fee of a working crew

Where profit loss is benefit loss when 1 pipe joint has not been welded. Standby fee is the cost for a welding crew standing by and doing nothing, including the workers' salary, insurance, meals, etc., and rental fee of all the equipment. To simplify the problem, we assumed the below conditions as per the cost investigation of pipe strings:

If a pipe delivered to the site couldn't be welded on-site, the pipe shall be returned to the stockyard, because the pipe would be not allowed to stock on site considering the safety requirement. And on average, the cost to stock the pipe on site will be $100 per piece. As per the estimated budget, $540 will be earned per one piece of pipe that welded on-site. Thus, the analysis could be done as the model of newsvendor model.

Given that Holding Cost: $h = 100$, Penalty Cost: $p = 540$, Pipe supply Demand of pipe string $D \sim N(19.61, 4.957^2)$.

As per the formulation of

$$S^* = F^{-1}\left(\frac{p}{p+h}\right) \tag{3}$$

where F is the CDF of $N(19.61, 4.957^2)$. Thus,

$$S^* = F^{-1}\left(\frac{540}{540+100}\right)$$

By using the function of NORMINV(0.84375, 19.61, 4.957) in Excel, we got that

$$S^* = 24.61$$

Thus, the optimal quantity of pipe supply is 25 pieces per day. According to Eq. 2, in which Z_α could be calculated as follow,

$$Z_\alpha = \frac{S^* - \mu}{\sigma} = \frac{25 - 19.61}{4.957} = 1.087$$

Thus, $\phi(Z_\alpha) = \phi(1.087) = 0.2396$. Accordingly, $g(S^*) = (100 + 540)(0.2396)(4.957) = 760.13$. On the other hand, if the project manager delivers pipes to the construction site at a mean quantity, 19 pieces per day, as the traditional method, the cost is going to be $g(S) = 1,587.22$. As per the percentage of cost-saving, it would be 52.11%.

5 DISCUSSION

As per the calculation in the last section, the result of demand calculated by the newsvendor model is higher than by the traditional method which is based on the average demand only, and the expected cost as per the traditional method is higher which indicates that we may be lost more benefit due to the backorder of the pipe supply in pipe string activity.

However, the holding cost is assumed as constant on daily basis and it would be more complicated in practice. The holding cost would be increasing if the rate couldn't be caught up by the welding team, then the demand will be lower than the result from the above analysis. So, the result would be the maximum demand requirement for pipe supply in pipe string activity.

It would be concluded that while we decide on how many resources should be invested in the activity, the relationship of HC and PC shall be taken into consideration. If $HC = PC$, we will face the 50-50 situation like gambling. Besides, if $PC > HC$, we may pay attention to penalty cost, and if $PC < HC$, we would pay more attention to cost control.

6 CONCLUSION AND OUTLOOK

Cost control is one of the important functions in a business; especially, in the business of large construction projects. This study investigated an operation of a gas pipeline construction project. The tradeoff between underage and overage pipe stringing activity was analyzed. Then, we made use of the newsvendor model to determine the optimal quantity. We found that the new method could save money for $827.09 per day, it accounted for up to 52.11%.

For further study, the relevant crew can set up other activities, such as the right of way cleaning and grading and right of way acquisition, based on the concept of this. Additionally, the team is also interested to apply other inventory policies, such as based stock policy, economic order quantity policy, etc., to activities in the construction project.

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